Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization

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Weakness, not generic compression alone, in shortcut-compatible learning problems

Research-Derived Experiments · preprint compiled from the project repository

Abstract

We study finite-domain model selection where training data admits both a local shortcut and a globally symmetry-compatible rule. Loss and in-distribution validation cannot distinguish the completions, and generic simplicity or flatness proxies can prefer the shortcut. We define **weakness** as the number of transformations under which a candidate function remains equivariant up to an output action. On cyclic and dihedral symbolic families (500 trials each), weakness recovers the invariant rule in 100% of trials (Wilson 95% lower bound 0.992) while the tested train-loss, validation, description-length, MDL-style, compression, and flatness proxies recover it in 0% (upper bound 0.008). A **data-inferred** selector, built from circular-domain transformation enumeration plus training-pair consistency, matches the oracle without using the hidden offset or reflection parameter. Across 256, 1024, and 4096 trained MLPs, learned-function weakness under the true group is the strongest correlate of OOD accuracy ($r = +0.817$, $+0.813$, $+0.8085$; 4096 CI $+0.798$ to $+0.819$), and an augmentation-fixed-effect check remains positive (residual $r = +0.488$). Vision gives the same ordering (Z_8 rotation, $r = +0.672$). Parity is a clean negative case, and S_n is a large-group boundary case: oracle weakness helps, but data-inferred group discovery degrades. Pythia-70M is reported only in an appendix as latent geometry, not behavioral evidence.

1. Introduction

Modern pipelines are often underspecified: many predictors have equivalent training or in-distribution validation performance but different deployment behavior. Shortcut learning is one familiar failure mode. Here we make the ambiguity explicit: the training data is compatible with both a local shortcut and a globally transportable rule. If the deployment distribution is generated by transformations, the relevant question is not only which hypothesis is shortest, but which hypothesis remains valid under those transformations. This paper tests Bennett-style *weakness* — compatible transformation volume — as a finite, measurable selection rule.

2. Weakness

Let $f : X \to X$ be a candidate function on a finite domain of size n, and let G act on X. We call f *compatible* with $g \in G$ if there exists $h \in G$ with $f(g\cdot x) = h\cdot f(x)$ for all x (with-action equivariance; this generalizes strict equivariance $h = g$). Weakness is the count of compatible group elements:

$$\mathbf{W_G(f) = | \{ g \in G : \exists\, h \in G, \forall x, f(g\cdot x) = h\cdot f(x) \} |.}$$

$W_G(f) = |G|$ means f is fully G-equivariant; $W_G(f) = 1$ means only the identity commutes. The *weakness selector* returns $\operatorname{argmax}_i W_{\hat{G}}(f_i)$ over a candidate pool, ties broken by the same compression proxy used by the baselines. Selection principle: when training constrains only part of a transformation orbit, train-consistent hypotheses compatible with more deployment-generating transformations should cover more of the missing orbit.

Tiny proposition: if deployment inputs are completed from observed examples by a subset S of G, then every compatible g in S transports observed labels along its orbit without contradiction.

More compatible transformations therefore cover more of the missing orbit, provided the candidate group is aligned with the deployment shift.

3. Symbolic separation

We build four task families, each with a known transformation group and a training distribution that admits both a train-perfect local shortcut and the true invariant: *cyclic_prefix_shift* (ℤ_n), *dihedral_reflection* (D_n), *parity_coset* (Z_2), and *color_permutation* (S_n). We compare eleven selectors over 500 trials per family with Wilson 95% intervals. Candidate pools are explicit: cyclic has K=n+1 (8/12/14 here), dihedral K=3, parity K=4, and color-permutation K=3..6 in the reported seeded run.

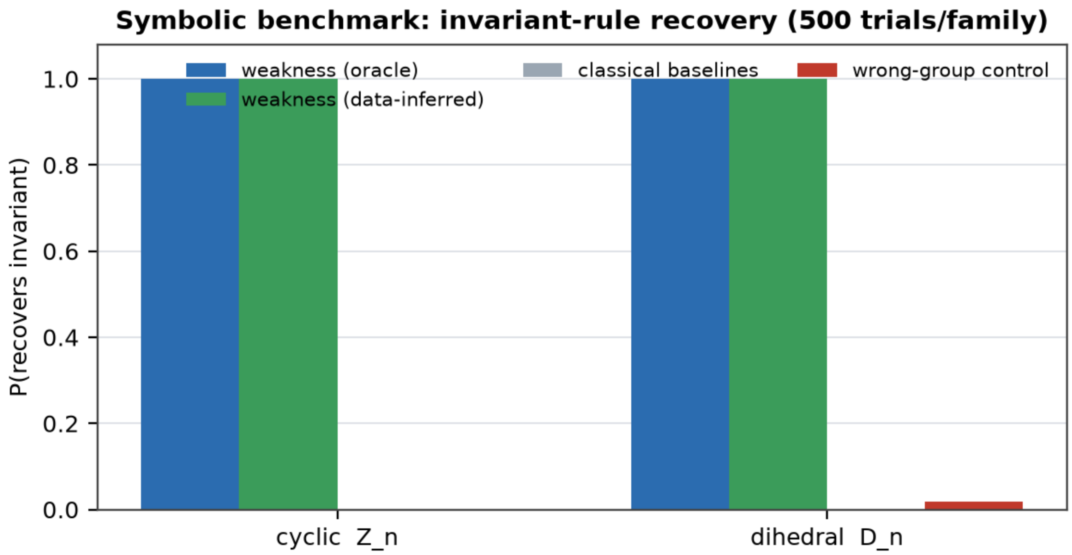


Figure 1. On cyclic and dihedral families, only weakness-based selectors recover the invariant rule (1.000; Wilson LB 0.992); training loss, simplicity, MDL, compression, flatness, and validation all recover it 0.000 (UB 0.008). The data-inferred selector uses circular-domain enumeration plus training-pair consistency, not the hidden offset/reflection, and matches the oracle. The wrong-group control is near zero.

Family	weakness (oracle)	data-inferred	classical baselines	wrong-group
cyclic Z_n	1.000	1.000	0.000	0.002
dihedral D_n	1.000	1.000	0.000	0.018
color S_n (partial)	0.824	0.138	0.000	0.000
parity Z_2 (negative)	0.000	0.000	0.000	0.038

Table 1. Invariant-recovery rate by selector and family. Cyclic and dihedral show a perfect, non-overlapping separation; S_n is a partial win; parity is a clean negative (|G|=2 too small to disambiguate).

4. Neural weakness predicts OOD accuracy

We train 256 small MLPs, replicate at 1024 models, and rescale to 4096 models via Modal on cyclic tasks with n in {7,11,13}, train window w in {2,3,4}, diverse depth, width, initialization, optimizer, learning rate, weight decay, and augmentation. From each model we extract the argmax function table and compute true-group weakness together with classical predictors, correlating each against held-out OOD accuracy.

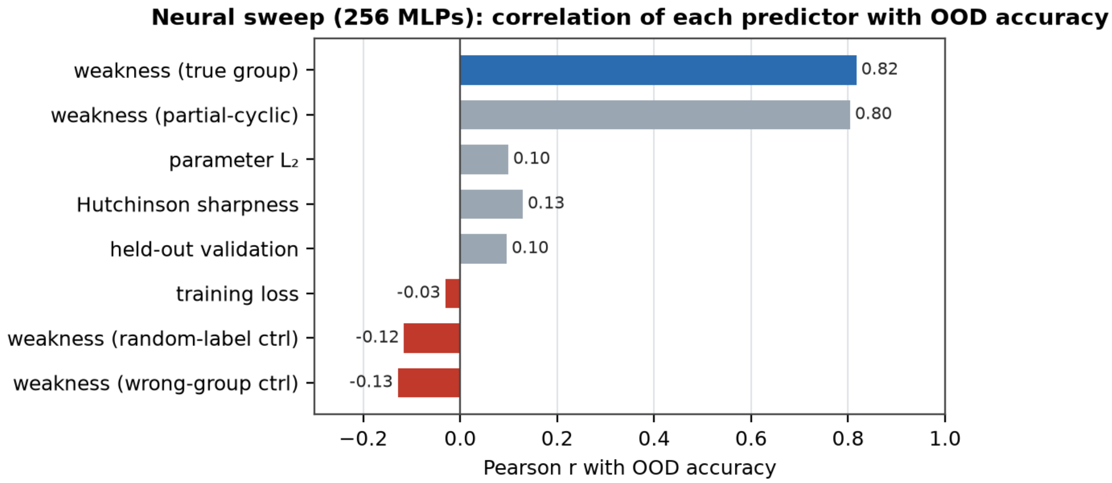


Figure 2. Across 256 MLPs, weakness under the true group is the single strongest predictor of OOD accuracy (Pearson r = +0.817 locally, +0.813 at 1024 models, +0.8085 at 4096 models). Training loss, validation, parameter L_2, and sharpness are weaker; wrong-group and random-label controls are correctly negative.

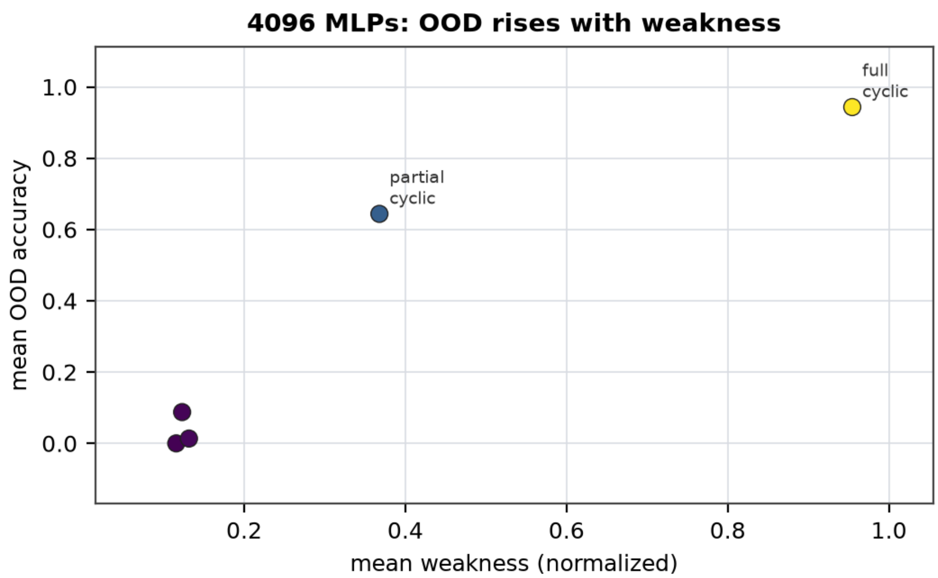


Figure 3. The per-augmentation gradient is monotone in weakness: full-cyclic (orbit completion) reaches 94% OOD and 0.95 weakness, while none / wrong-reflection collapse to ≤1.4% OOD and ≤0.13 weakness. Augmentations that approximately respect the symmetry raise weakness and OOD together; after augmentation fixed effects, residual r remains +0.488.

5. Scaling: vision

On a synthetic Z_8 rotated-stroke task (eight classes, 16×16, three of eight angles shown in training; a re-implementation of Perin and Deny's partial-orbit setup), weakness under the rotation group is again the dominant predictor across 96 models (r = +0.67), beating every classical predictor by ≥2×, with the pixel-permutation wrong-group control correctly anti-correlated (r = −0.34).

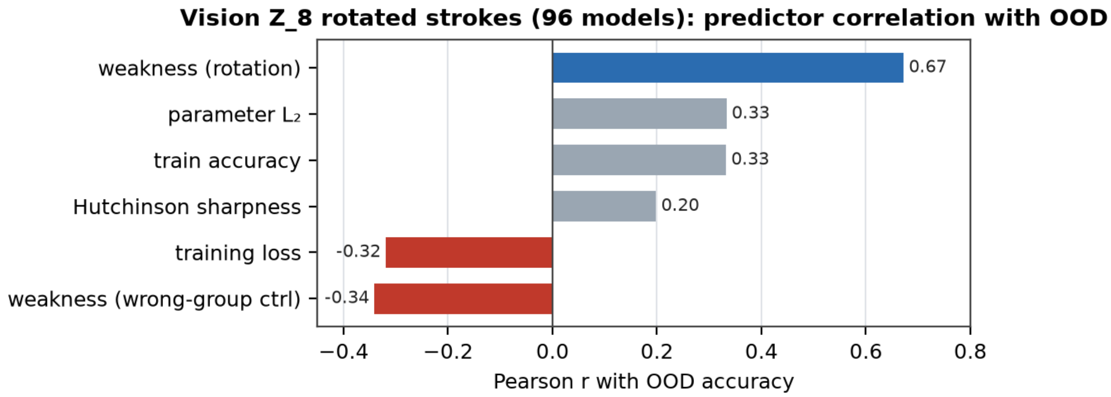


Figure 4. Vision (Z_8 rotation): weakness leads; the wrong-group (pixel-shuffle) control is anti-correlated, as expected of a model that has actually learned the rotation symmetry.

The language probe is moved to Appendix C and treated as exploratory latent geometry only; it is not used as behavioral OOD evidence.

6. Operating regime and limitations

Weakness ceases to discriminate when the candidate group is too small (parity, $|G|=2$) or too large/uninformative (full symmetric group, where wrong involutions have comparable centralizer-orbit sizes). This is a precise statement of when symmetry-volume is load-bearing, not a defect. Present limitations: small finite domains ($n \le 13$ symbolic, 16×16 vision); intentionally simple MDL/compression/completion-volume and sharpness proxies; the language latent→behavior chain unconfirmed; transformation discovery is still enumerative. The natural next steps are stronger compression/PAC-Bayes baselines, learned transformation generators, and training-time weakness regularization.

7. Discussion

The discriminating quantity between local shortcut and globally invariant rule — on families where both are train-perfect — is symmetry-compatible-hypothesis volume: a measurable, intervention-friendly, reparameterization-invariant quantity, not a parameter-space artifact. It operationalizes Bennett's weakness on neural function tables and answers the practical question of which heuristic to trust when training loss is tied. The safe thesis is not that compression is wrong; it is that, in these shortcut-compatible symmetry tasks, the relevant compression is compatibility with the transformations that generate the missing cases.

Appendix A. Methods and statistics

Symbolic families are exact finite tasks. Cyclic uses X=Z_n, n in {7,11,13}, truth f_b(x)=x+b mod n, a prefix train window, and a suffix OOD set. Dihedral uses X=Z_n, n in {7,9,11,13}, truth f_b(x)=b-x mod n, and D_n rotations/reflections. Parity uses an even domain and f(x)=x xor 1 with one parity coset held out. Color-permutation uses S_n, n in {4,5,6}, a sampled non-identity permutation, sparse training inputs, and unobserved inputs as OOD.

Selector	Score	Tie-break	Group access
train_loss	train accuracy	shorter form	none
validation	leave-one-observed-pair accuracy	shorter form	none
simplicity	shorter form_length	random exact tie	none
compression	form_length + 20*train_errors	random exact tie	none

mdl_program	2^-form_length	shorter form	none
flatness_proxy	completion-volume proxy, not Hessian flatness	shorter form	none
weakness_oracle	W_G(f)	compression	true task group
weakness_wrong_group	W under random perms	compression	wrong control
weakness_noisy_group	W under noisy subset	compression	noisy true group
weakness_data_inferred	W under inferred G_hat	compression	train pairs only
random	uniform train-perfect candidate	-	none

Appendix Table A1. Exact selector definitions. Classical baselines are the first six rows; validation is in-distribution only.

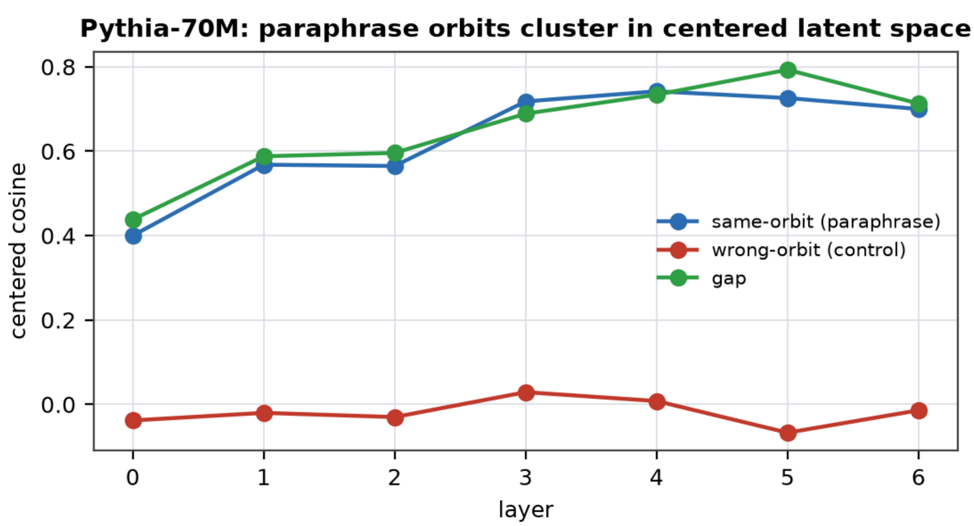
Data-inferred group discovery enumerates a family-specific domain prior (Z_n shifts for cyclic, D_n rotations/reflections for dihedral, parity swap for parity, and a small cyclic heuristic for S_n), then keeps every input-side transformation g for which some output-side transformation h is non-contradictory on observed training pairs. It uses training inputs, training outputs, and the domain prior only: no test labels, hidden offset, hidden reflection, sampled permutation, or invariant-candidate identity.

Statistic	Value
4096 true-group Pearson r	+0.8085 (95% CI +0.7976 to +0.8189)
4096 partial-cyclic Pearson r	+0.7940 (95% CI +0.7824 to +0.8051)
4096 validation Pearson r	+0.0924 (95% CI +0.0620 to +0.1227)
4096 parameter L2 Pearson r	+0.2533 (95% CI +0.2244 to +0.2818)
4096 fixed-effect residual r	+0.4883 (95% CI +0.4647 to +0.5113)
Fixed-effect beta	+0.3652 (95% CI +0.3452 to +0.3853)

Appendix Table A2. Fisher-z intervals and augmentation fixed-effect check. The headline effect is partly augmentation-mediated, but a within-condition learned-function signal remains.

Appendix C. Exploratory language check

Pythia-70M hidden states for 24 concepts x 3 paraphrases show centered same-orbit cosine gaps of +0.44 to +0.79 after All-but-the-Top centering, but per-concept next-token behavioral consistency is not predicted at this scale (Pearson range -0.35 to -0.13, N=24). This supports only a latent-geometry statement: paraphrase orbits cluster in centered representation space.



Appendix Figure C1. Pythia-70M paraphrase orbits cluster in centered latent space (gap peaks at +0.79, layer 5). This is a latent-geometry result only.

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